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RESEARCH ARTICLE

Improvement of Adaptive Cruise Control System Performance on Sloped Roads Based on Adaptive Neuro-Fuzzy Inference System

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ABSTRACT Adaptive cruise control is a system that maintains an appropriate distance from vehicles ahead while driving. Although Adaptive cruise control systems operate stably on flat roads, they can become unstable on sloped roads. This study presents a method to enhance the performance of Adaptive cruise control systems on sloped roads using fuzzy logic control techniques and the adaptive neuro-fuzzy inference system. First, an Adaptive cruise control system is described based on state-space equations. Then, a fuzzy logic controller is designed and applied to the Adaptive cruise control system, and improved Adaptive cruise control performance is verified via computer simulations. The two inputs of the fuzzy logic controller are the speed error and the distance between vehicles. The output is the improved control force. In addition, to achieve enhanced performance under varying road conditions, the adaptive neuro-fuzzy inference system is applied. The adaptive neuro-fuzzy inference system is designed by training the designed fuzzy logic controller with data extracted from scenarios in which the road gradient angle changes. Simulations are performed by applying the adaptive neuro-fuzzy inference system applied to the Adaptive cruise control system. These simulations demonstrate that improved results are obtained even under new road conditions. This paper demonstrates that the design of the adaptive neuro-fuzzy inference system controller can effectively enhance the performance of the Adaptive cruise control system on sloped roads.

INDEX TERMS Adaptive cruise control (ACC), fuzzy logic control, fuzzy inference system (FIS), adaptive neuro-fuzzy inference system (ANFIS).

I. INTRODUCTION

Adaptive cruise control (ACC) systems are driver assistance systems that automatically control vehicle speed to maintain a safe distance from the lead vehicle while driving [1]. Since this concept was initially introduced and applied to vehicles, ACC systems have evolved continuously [2]. ACC systems help reduce driver fatigue and improve driving convenience [3]. Moreover, as a crucial technology to achieve full automation, research is being conducted to improve the performance of ACC systems [4], [5], [6].

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ACC systems function effectively on flat roads but may experience performance degradation on sloped roads [7]. Many accidents occur when ACC systems fail to control vehicle speed on sloped roads [8]. Therefore, enhancing the stability of ACC systems on sloped roads is crucial. Ongoing research aims to improve the performance of ACC systems on sloped roads [9], [10].

To enhance the performance of ACC systems on sloped roads, James proposed an adaptive feedback control based on large-scale experimental data [9]. By extracting appropriate control signals according to various road angles, an algorithm was designed to efficiently control vehicle speed on sloped roads. The dynamic changes in the vehicle on sloped roads are reflected in real time by feedback control. However, this

method is dependent on experimental data; thus, errors may occur in new road environments. In addition, due to complex computations, unexpected actions may be executed if the system response is delayed.

Liang proposed an optimal control that considers both string stability and vehicle speed on sloped roads [10]. String stability indicates that variations in the speed of the lead vehicle are not amplified when transmitted to the following vehicle. By predicting the vehicle speed on sloped roads and adjusting the distance, the system maintains driving stability. However, maintaining string stability while optimizing vehicle speed and distance requires processing a large amount of data in real time. Therefore, the system may not react immediately on roads with rapidly changing slopes. In addition, the frequent acceleration and deceleration required by the optimal control algorithm for fuel saving can degrade ride comfort.

This study presents an application of fuzzy logic control and the adaptive neuro-fuzzy inference system (ANFIS) to minimize complex computations and data dependencies. Fuzzy logic control, presented by Zadeh [11], [12], applies fuzzy logic principles in the control domain. By defining input and output variables as fuzzy sets and setting up the rule base, fuzzy logic control derives the desired output in complex situations [13]. The defuzzification process converts uncertain data into a specific output [14]. In other words, fuzzy logic control provides a control algorithm that can flexibly respond to uncertain and complex situations [15], [16], [17]. Recent studies have focused on enhancing the performance and stability of fuzzy systems [18], [19], [20], [21], [22]. This study aims to apply fuzzy logic control techniques to maintain a stable distance from a leading vehicle on sloped roads.

An ANFIS combines the reasoning ability of fuzzy logic with the learning ability of neural networks [23]. This system was proposed by Jang [24], [25], [26]. By learning the relationships between input and output data, the ANFIS enhances prediction accuracy for complex problems and generates a clear rule-based system through training [27]. Recent studies have applied the ANFIS to various fields [28], [29], [30]. The ANFIS enhances the performance of ACC systems by providing improved control forces even under changing road conditions. To the best knowledge of the authors, applying the ANFIS to ACC systems is the first approach thus far.

To improve the performance of ACC systems on sloped roads, a fuzzy logic controller is designed. The two input variables are distance and speed error. The output variable is an improved control force. A rule base is designed based on the input and output variables. The fuzzy logic controller is applied to the ACC system to verify the improved performance. Data are extracted through simulations under changing road conditions. The fuzzy logic controller is trained on the data to design the ANFIS. With this approach, the improved ACC system achieves high accuracy

and demonstrates superior performance under new road conditions.

The ANFIS controller learns from training data to maintain stable performance even under changing road conditions. It overcomes the limitations of adaptive feedback control, which relies on experimental data and performs complex computations in real-time. Because ANFIS is based on learning from training data instead of complex real-time computations, it also addresses the limitations of optimal control, which requires processing large amounts of data in real time and causes frequent acceleration and deceleration. In summary, the proposed ANFIS control method enhances the performance of ACC systems in dynamic road environments by overcoming the difficulties associated with existing methods that are dependent on experimental data and complex real-time computations.

In Section II, an ACC system is implemented to observe the distance between vehicles and the control force on sloped roads. In Section III, a fuzzy logic controller is designed and applied to the ACC system to verify the improved performance. Section IV presents the limitations of the fuzzy logic controller, and the ANFIS is proposed as a solution to overcome these limitations of the fuzzy logic controller.

II. MODELING OF ADAPTIVE CRUISE CONTROL SYSTEMS

A. SUMMARY OF ADAPTIVE CRUISE CONTROL SYSTEMS

In this study, the ACC system model proposed by Ulsoy is adopted [31]. The physical constants, vehicle parameters, and state-space equations are also referenced. Lead and controlled vehicles are introduced to apply the ACC system. The state-space equations are based on the mass, drag coefficient, frontal area, and rolling resistance. The ACC system is established based on these equations. This process is performed in Matlab/Simulink[®].

TABLE 1. Physical constants.

Parameter	Symbol	Value	Unit
Gravitational acceleration	g	9.81	m/s ²
Air density	ρ	1.202	kg/m ³
Initial speed	u_0	30.0	m/s
Wind speed	u_w	0.0	m/s
Road slope angle	Θ	0.0	radian
Rate of change of road slope angle	Θ'	0.0	radian/s

TABLE 2. Vehicle parameters.

Parameter	Symbol	Control Vehicle	Lead Vehicle
Mass	m	1000 kg	1500 kg
Drag coefficient	C_d	0.5	0.6
Frontal area	A_f	1.5 m ²	1.95 m ²
Rolling resistance	f	0.015	0.015

1) OPEN-LOOP SYSTEM

The state-space equations control the distance between vehicles and the speed of the controlled vehicle. These

equations include the control input and state variables, which represent the speed and distance between vehicles.

$$\dot{x}_1 = x_3 - x_2, \quad (1)$$

$$\dot{x}_2 = -\frac{1}{\tau_c}x_2 + \frac{K_c}{\tau_c}u + \frac{K_c}{\tau_c}w_c, \quad (2)$$

$$\dot{x}_3 = -\frac{1}{\tau_l}x_3 + \frac{K_l}{\tau_l}w_l. \quad (3)$$

where x_1 denotes the distance between vehicles, x_2 denotes the speed of the controlled vehicle, and x_3 denotes the speed of the lead vehicle; u denotes the control force applied to the controlled vehicle; w_c and w_l represent disturbances to the controlled and lead vehicles, respectively [15] and [17]; which represent forces with a gradient; K and τ are calculated as follows:

$$K = \frac{1}{\rho C_d A_r (u_0 + u_w)}, \quad (4)$$

$$\tau = m \cdot K. \quad (5)$$

where K represents the aerodynamic coefficient of vehicles and reflects vehicle air resistance; K_c and K_l denote the aerodynamic coefficients of the controlled and lead vehicles, respectively; τ denotes the time constant of vehicles which is used to define the dynamic characteristics of vehicles and the characteristics of the system; τ_c and τ_l denote the time constants of the controlled and lead vehicles, respectively.

2) DESIGN OF POLE PLACEMENT CONTROLLER

To achieve the desired system response, a state feedback controller is proposed using the pole placement method.

The desired poles are set considering the stability and performance of the system. The poles and state feedback gain matrix are as follows:

$$p_c = \begin{bmatrix} -0.36 + 0.36i \\ -0.36 - 0.36i \\ -1.08 \\ -1.18 \end{bmatrix}, \quad (6)$$

$$k = [-3061.6 \ 2952.955 \ -1279.168 \ -203.904]. \quad (7)$$

3) CLOSED-LOOP SYSTEM

A closed-loop system is implemented using the state feedback controller. The state-space equations are as follows:

$$\dot{x}_1 = x_5 - x_2, \quad (8)$$

$$\dot{x}_2 = -\frac{1}{\tau_c}x_2 + \frac{K_c}{\tau_c}u + \frac{K_c}{\tau_c}w_c, \quad (9)$$

$$\dot{x}_3 = x_1 - r, \quad (10)$$

$$\dot{x}_4 = x_3, \quad (11)$$

$$\dot{x}_5 = -\frac{1}{\tau_l}x_5 + \frac{K_l}{\tau_l}w_l, \quad (12)$$

$$u = -k(1)x_1 - k(2)x_2 - k(3)x_3 - k(4)x_4. \quad (13)$$

where x_1 denotes the distance between vehicles, x_2 denotes the speed of the controlled vehicle, x_3 denotes the integral of the distance error, x_4 denotes the double integral of the distance error, x_5 denotes the speed of the lead vehicle, and u denotes the control force applied to the controlled vehicle. This control force regulates the speed of the controlled

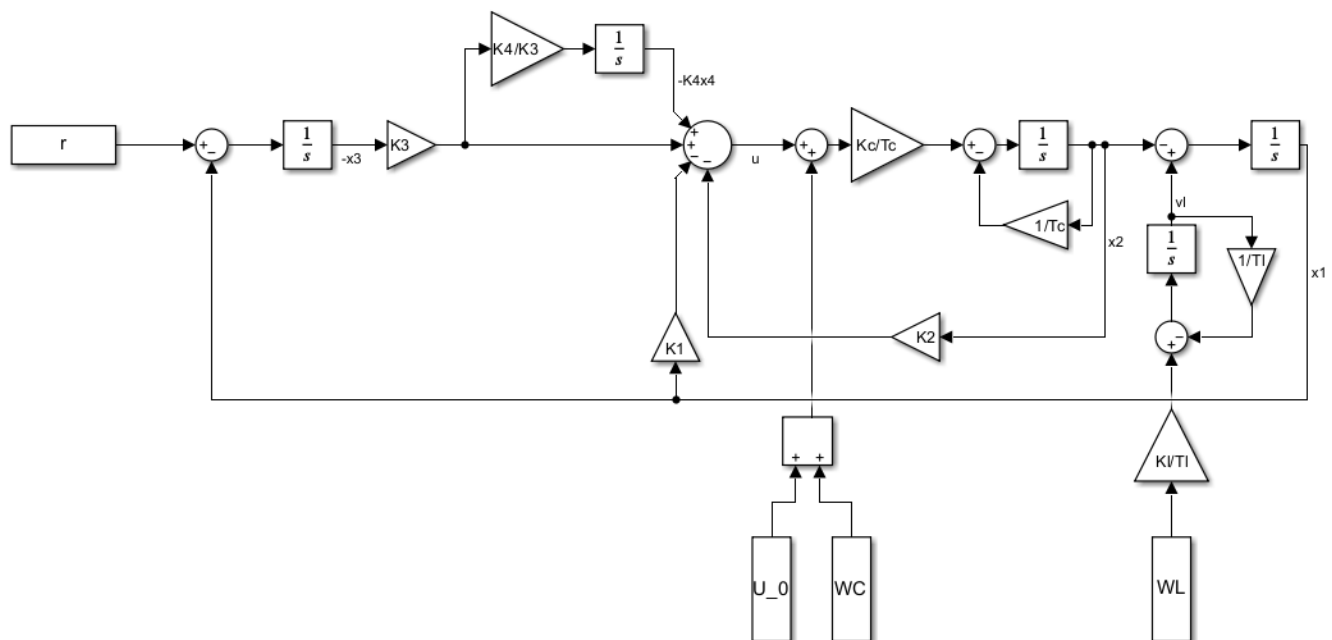


FIGURE 1. Block diagram of the ACC system. This figure shows the block diagram implementation of the state-space equations for the closed-loop system (8), (9), (10), (11), (12), (13). r is a reference input to maintain the desired distance. U_0 denotes the initial control force. w_c and w_l denote disturbance inputs for the controlled and lead vehicles, respectively. u is the control force that is calculated to maintain the desired distance. The simulation results for distance, speed, and control force are presented in Figure 2.

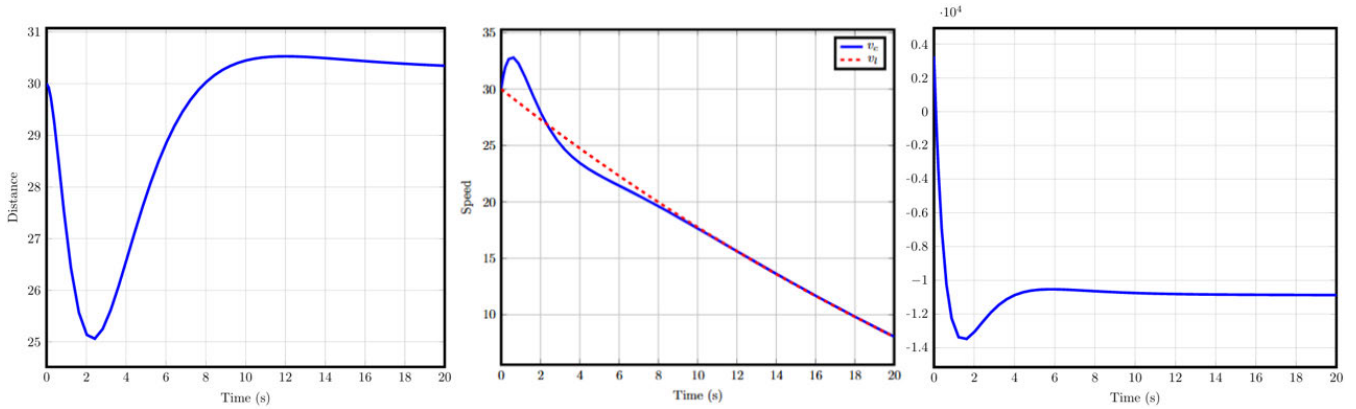


FIGURE 2. Graphs of the ACC system without a controller on sloped roads. The first graph shows the distance between the two vehicles on a sloped road. When two vehicles initially enter a downhill road, they become closer than the set distance. The second graph shows the speeds of lead and controlled vehicles. The solid and dashed lines represent the speeds of the controlled and lead vehicles, respectively. The speed of the lead vehicle gradually decreases from 30m/s. The speed of the controlled vehicle is adjusted with the control force to maintain a consistent distance between the two vehicles. The third graph shows the control force applied to the controlled vehicle. The control force adjusts the speed of the controlled vehicle to maintain the set distance between the two vehicles.

vehicle. r denotes the reference input representing the desired distance to be maintained, as determined by the user. Figure 1 shows a Matlab/Simulink[®] block diagram implementation of the state-space equations for the closed-loop system.

The reference input, disturbance input, and initial state vector are as follows:

$$r = 30, \quad (14)$$

$$w_c = m_c g (f_c \sin(\Theta) - \cos(\Theta)) \Theta', \quad (15)$$

$$U_0 = \frac{u_0}{K_c}, \quad (16)$$

$$w_l = -800(1 + 0.01t), \quad (17)$$

$$x_{c0} = \begin{bmatrix} 30 \\ u_0 \\ 0 \\ \frac{u_0}{K_c} + k(1) \cdot 30 + k(2) \cdot u_0 \\ \frac{k(4)}{u_0} \end{bmatrix}. \quad (18)$$

where r represents a reference input indicating the target distance; m_c denotes the mass of the controlled vehicle; f_c denotes the rolling resistance of the controlled vehicle; U_0 denotes the initial control force vector, which is the force applied to the vehicle at the beginning of the simulation; x_{c0} denotes the initial state vector, which represents the initial values of x_1 , x_2 , x_3 , x_4 , and x_5 ; x_{c0} denotes the state of the system at the beginning of the simulation.

B. IMPLEMENTATION OF ADAPTIVE CRUISE CONTROL SYSTEMS

1) BLOCK DIAGRAM OF THE ACC SYSTEM

In this study, a block diagram is designed based on the state-space equations(8), (9), (10), (11), (12), and (13). This block diagram is designed in Matlab/Simulink[®]. Figure 1 shows the block diagram of the ACC system. Here, r represents a reference input to maintain the desired distance(14); U_0 denotes the initial control force(16); w_c and w_l denote

the disturbance inputs for the controlled and lead vehicles, respectively(15) and (17); θ and θ' are set to -0.3 and -1 , respectively, and are applied to w_c to simulate a sloped road.

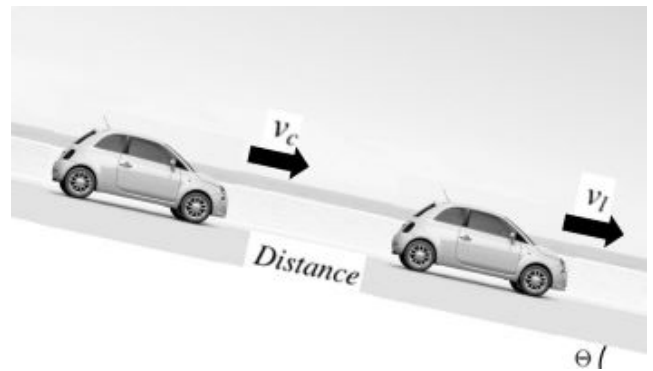


FIGURE 3. Illustration of the ACC system on a sloped road. This figure shows a simulation of the ACC system on a downhill road. The speeds of the lead and controlled vehicles are represented by v_l and v_c , respectively, and the downhill slope angle is represented by θ .

2) SIMULATION ON SLOPED ROADS

θ and θ' are set to verify simulation results on sloped roads. In this study, a downhill road with negative θ and θ' are considered. θ and θ' are set to -0.3 and -1 , respectively, for the simulation. The simulation is performed for 20 seconds. Figure 2 shows the performance of the ACC system on sloped roads.

The controlled vehicle adjusts its speed to maintain a consistent distance from the lead vehicle. The control force adjusts the speed of the controlled vehicle. The first graph shows the distance between the lead and controlled vehicles. The minimum distance between the two vehicles is 25m. However, the user-defined target distance is 30m. There is an error of 5m from the desired value, and reducing this error is the main objective of the controller design. The

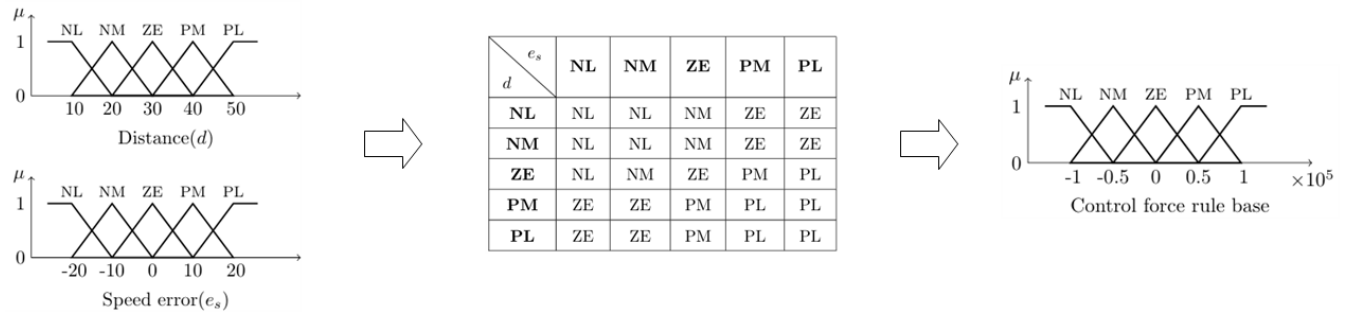


FIGURE 4. Visualization of fuzzy logic control. This figure illustrates the process of fuzzy logic control. Here, NL, NM, ZE, PM, and PL represent negative large, negative medium, zero, positive medium, and positive large, respectively. The two inputs are the distance and speed error. The speed error is the difference between the speeds of lead and controlled vehicles (19). The output is the control force. The inputs and output are fuzzified using triangular membership functions. A rule base is established to achieve the desired results. The output is defuzzified using the Mamdani inference method to obtain the control force.

second graph illustrates the speeds of the lead and controlled vehicles. The solid and dashed lines indicate the speeds of the controlled and lead vehicles, respectively. The speed error between the two vehicles occurs during the initial six seconds of the simulation. This indicates that the distance between the lead and controlled vehicles temporarily increases or decreases. This phenomenon needs to be improved through the controller design. The third graph illustrates the control force applied to the controlled vehicle. The control force adjusts the speed of the controlled vehicle to maintain a constant distance from the lead vehicle. The control force should be adjusted to improve the performance of the ACC system on sloped roads.

III. DESIGN OF FUZZY LOGIC CONTROLLER

A fuzzy logic controller is designed to improve the performance of the ACC system on sloped roads. The design process of the fuzzy logic controller is described as follows.

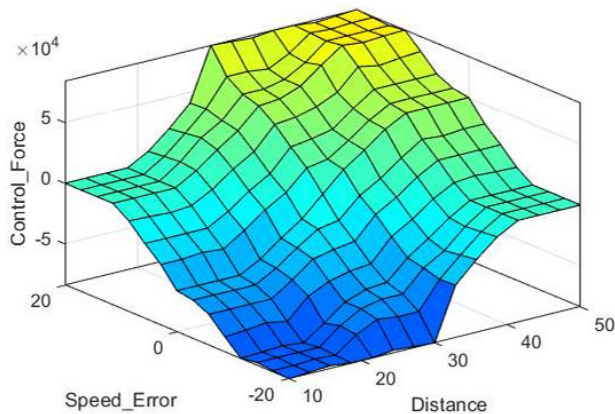


FIGURE 5. Surface of the fuzzy logic controller. This figure shows the surface of the designed fuzzy logic controller. This three-dimensional (3D) graph shows the input and output variables and their ranges. This graph also illustrates the output value for a specific input. This graph provides a clear view of the operation of the FIS.

Figure 5 shows a 3D graph that illustrates the surface of the designed fuzzy logic controller. This graph shows the relationship between the input and output variables of the FIS.

A. INPUT AND OUTPUT VARIABLES

The input variables of the fuzzy logic controller should reflect the current state of the vehicle and environmental conditions. Therefore, the speed error and distance between the vehicles are set as input variables. The speed error is the difference between the speeds of the lead and controlled vehicles (19). The output variable determines the behavior of the controlled vehicle. Therefore, the control force is set as the output variable.

$$e_s = v_l - v_c. \quad (19)$$

B. FUZZY SETS AND MEMBERSHIP FUNCTIONS

The input and output variables are divided into fuzzy sets of NL, NM, ZE, PM, and PL, which represent negative large, negative medium, zero, positive medium, and positive large, respectively. The fuzzy sets are represented through triangular membership functions.

C. FUZZY INFERENCE SYSTEM AND RULE BASE

The Mamdani inference method [32] is applied to the fuzzy inference system in this section. In the Mamdani inference method, the output is represented as fuzzy sets. The rule base establishes the relationship between inputs and outputs [33].

D. SIMULATION OF FUZZY LOGIC CONTROLLER

The designed fuzzy logic controller is applied to the ACC system for simulation. Figure 6 shows a block diagram of the ACC system with the fuzzy logic controller. The input variables of the fuzzy logic controller are the speed error and the distance between the two vehicles. The output of the fuzzy logic controller is an improved control force. The improved control force is combined with the existing control force, u , to generate a new control force. The new control force adjusts the speed of the controlled vehicle to maintain the desired

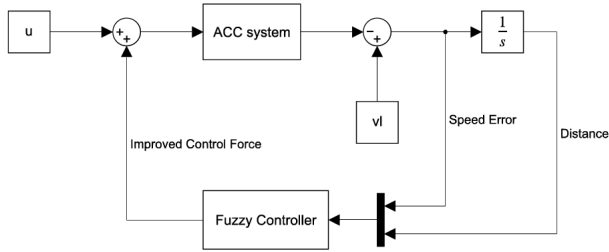


FIGURE 6. The ACC system with designed fuzzy logic controller. This figure shows how the fuzzy logic controller is applied in the ACC system. u and v_l denote the control input and the speed of the lead vehicle. Inputs of the fuzzy logic controller are the speed error and the distance between two vehicles. Output of the fuzzy logic controller is an improved control force.

distance between the two vehicles. The simulation results are presented in Figure 7.

Figure 7 shows the performance of the ACC system on a sloped road when the fuzzy logic controller is applied. The first graph shows the distance between the two vehicles. The solid and dashed lines represent the simulation results with and without the fuzzy logic controller, respectively. The ACC system with the fuzzy logic controller maintains a distance that is closer to the user-defined value ($r = 30m$). The second graph shows the speed of each vehicle when the fuzzy logic controller is applied. The solid and dashed lines represent the speeds of the controlled and lead vehicles, respectively. Compared to the second graph in Figure 2, the speed of the controlled vehicle is shown to be significantly closer to that of the lead vehicle. This indicates that the two vehicles maintain a consistent distance when the fuzzy logic controller is applied. The third graph shows an improved control force. The improved control force is the output value based on the rule base of the fuzzy logic controller. It is added to the existing control force to adjust the speed of the controlled vehicle. The fuzzy logic controller outputs a negative control force to regulate the increasing speed of the controlled vehicle on downhill roads.

IV. ADAPTIVE NEURO-FUZZY INFERENCE SYSTEM

A. LIMITATION OF THE FUZZY LOGIC CONTROLLER

The ACC system is implemented on a road with specific road slopes. An improved result is obtained by designing a fuzzy logic controller suitable for road slopes. However, a fuzzy controller that is specifically designed for certain road slopes may not perform effectively on varying slopes.

The first graph in Figure 8 shows the simulation results of the fuzzy logic controller under different road conditions. The fuzzy logic controller shown in Figure 5 is applied. The solid, dashed, and dotted lines represent the performance of the fuzzy logic controller when $\theta' = -1, -2$, and -4 , respectively, when applied to the ACC system. As θ' decreases, the performance of the ACC system with the fuzzy logic controller decreases. The first graph in Figure 8 highlights the need for a new controller design for a new road condition.

The second graph in Figure 8 presents the simulation results of three fuzzy logic controllers under different road conditions. Figure 9 shows the three fuzzy logic controllers designed for specific road conditions. Figure 9(a), (b), and (c) show the fuzzy logic controllers designed for $\theta' = -1, -2$, and -4 , respectively. The solid, dashed, and dotted lines represent the performance of controllers (a), (b), and (c), respectively, when applied to the ACC system. When compared to the first graph, the distance error in the second graph is noticeably smaller. This is because a controller specifically designed for each road slope is applied to the ACC system. As the road slope increases, the vehicle speed also increases. When the vehicle speed increases, a greater control force is required to control the increasing speed. In other words, a new controller should be applied depending on the road angle. However, it is not practical to change the controller for the ACC system each time the road conditions change. To overcome this limitation of the fuzzy logic controller, this study proposes the application of the adaptive neuro-fuzzy inference system (ANFIS).

B. CONCEPT OF ANFIS

The ANFIS is a hybrid intelligent system that combines the learning capabilities of artificial neural networks and the interpretation capabilities of FIS. The ANFIS is based on Sugeno's method [34]. Sugeno's method is a fuzzy inference system in which the output is represented as a constant or a linear function. The output of Sugeno's method is given by (20).

$$R_k = p_k x + q_k y + r_k. \quad (20)$$

where R_k represents the output of the k -th rule; p_k , q_k , and r_k are linear parameters associated with the k -th rule; x and y are input variables. The ANFIS processes large amounts of data through neural networks and optimizes the system through iterative training, which requires simple and fast computations. Sugeno's method does not require complex defuzzification and allows for easy application of optimization algorithms such as gradient descent [35]. The reason why the ANFIS is based on Sugeno's method.

The ANFIS structure typically consists of five layers [36], [37]. Figure 10 shows an example of the ANFIS architecture with inputs x and y . Figure 10 explains the ANFIS through each layer. The functions of each layer are explained, along with their corresponding equations.

1) LAYER 1: FUZZIFICATION LAYER

Each node in this layer represents a fuzzy membership function that calculates the degree to which the input variables belong to a given fuzzy set. If x and y are two input variables, the output of the node in the first layer is calculated as follows.

$$O_{1,i} = \mu_{A_i}(x), \quad O_{1,j} = \mu_{B_j}(y). \quad (21)$$

where $O_{1,i}$ and $O_{1,j}$ denote the outputs of the node in layer 1; $O_{1,i}$ and $O_{1,j}$ denote the degree of membership of inputs in a

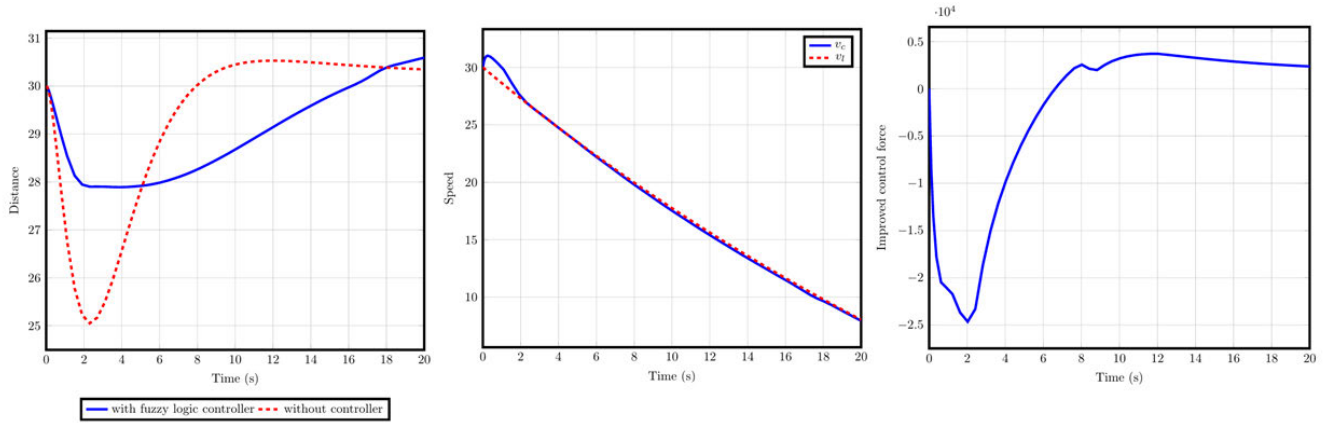


FIGURE 7. Graph of the ACC system with the fuzzy logic controller. This figure illustrates the performance of the ACC system with the fuzzy logic controller. The first graph shows the distance between the two vehicles. The solid and dashed lines represent the performance of the ACC system with and without the fuzzy logic controller, respectively. This graph shows how the fuzzy logic controller effectively improves the performance of the ACC system on sloped roads. The second graph shows the speeds of the two vehicles. The solid and dashed lines represent the speeds of the controlled and lead vehicles, respectively. Compared with the second graph in Figure 2, the speed error between the two vehicles decreases. This shows that the fuzzy logic controller effectively controls the speed of the controlled vehicle through the control forces. The third graph shows an improved control force, which is the output of the fuzzy logic controller. The improved control force is added to the existing control force to control the speed of the controlled vehicle and maintain a constant distance from the lead vehicle on sloped roads.

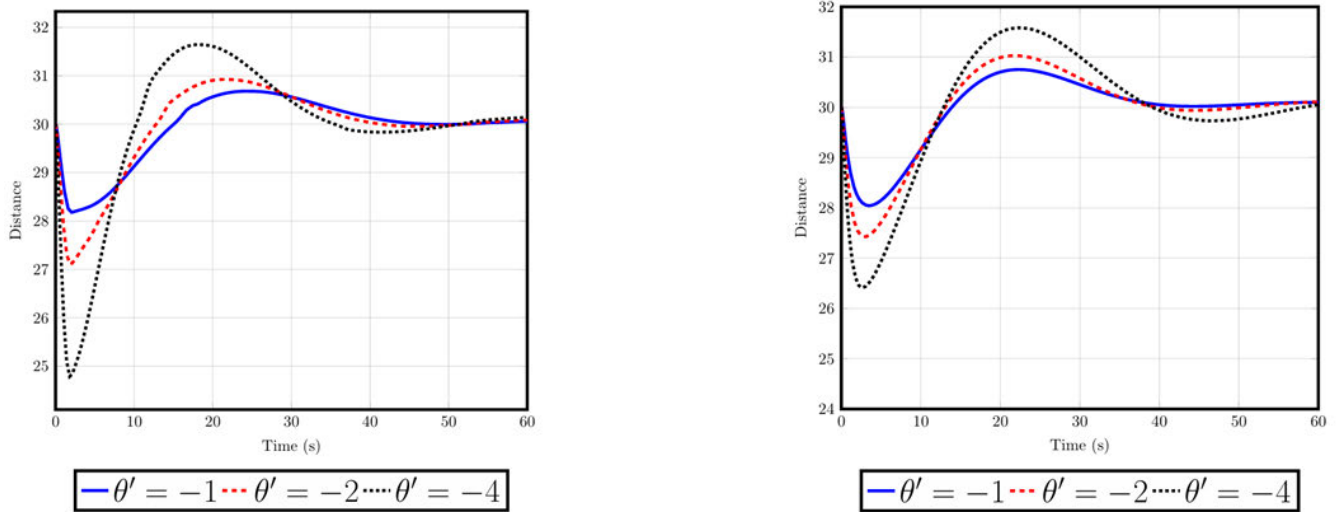


FIGURE 8. Performance of fuzzy logic controllers. The first graph shows the simulation results of the fuzzy logic controller under different road conditions. The fuzzy logic controller shown in Figure 5 is applied. The solid, dashed, and dotted lines represent the performance of the fuzzy logic controller when $\theta' = -1$, -2 , and -4 , respectively. As θ' decreases, the performance of the ACC system with the fuzzy logic controller degrades. This graph highlights the need for a new controller design for a new road condition. The second graph shows the performance of the fuzzy logic controllers shown in Figure 9 designed for specific angles. The solid, dashed, dotted lines represent the performance of fuzzy logic controllers designed for road conditions with θ' values of -1 , -2 , and -4 , respectively. These simulations are performed under road conditions with θ' values of -1 , -2 , and -4 . Compared to the first graph, the distance error appears small in the second graph. This is because a controller designed for each road slope is applied to the ACC system.

specific fuzzy set; $\mu_{A_i}(x)$ and $\mu_{B_j}(y)$ denote the membership functions for inputs x and y , respectively.

2) LAYER 2: RULE LAYER

This layer establishes fuzzy rules. Each node calculates the strength of the rule by multiplying the input membership values. In other words, this node can be viewed as executing an AND operation.

$$O_{2,k} = w_k = \mu_{A_i}(x) \cdot \mu_{B_j}(y). \quad (22)$$

where $O_{2,k}$ denotes the output of the k -th node in layer 2; w_k represents the output of the k -th rule, which is the weight of the rule.

3) LAYER 3: NORMALIZATION LAYER

In this layer, the weight of each rule is normalized. Weight normalization is necessary to represent how much the output of each rule contributes to the overall output, normalization is performed by dividing the weight of each rule by the sum

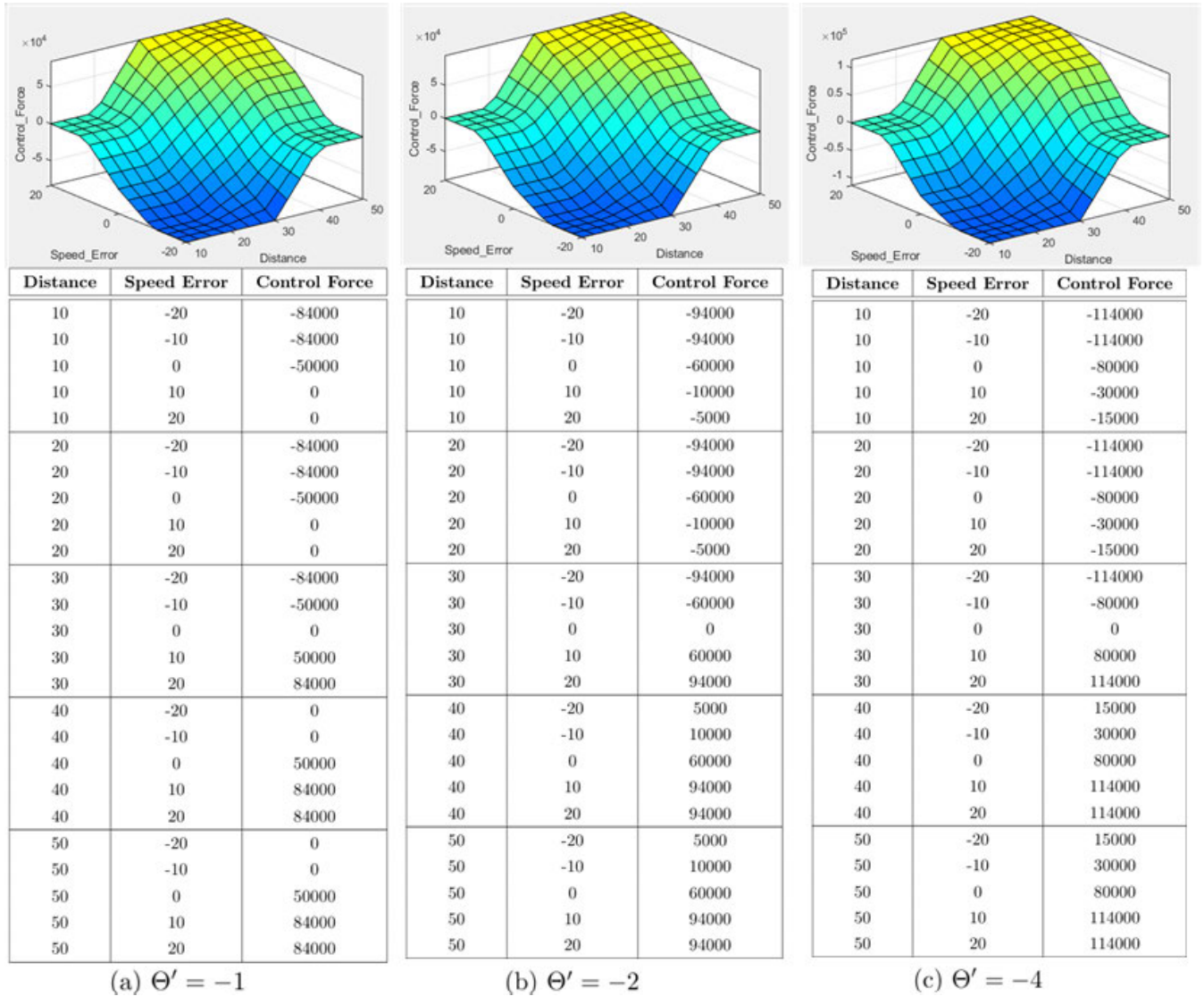


FIGURE 9. Surface of the fuzzy logic controller at various road slope angles. This Figure shows the input and output data of fuzzy logic controllers designed for specific road angles. (a), (b), and (c) are fuzzy logic controllers designed for road angles with θ' values of -1, -2, and -4, respectively. As the road slope increases, the vehicle speed increases. When the vehicle speed increases, a greater control force is required to control the increasing speed. Therefore, fuzzy logic controllers are designed to generate larger control forces as θ' value increases. The table shows the control force values corresponding to input values.

of the weights of all rules.

$$O_{3,k} = \bar{w}_k = \frac{w_k}{\sum_{i=1}^N w_i}. \quad (23)$$

where $O_{3,k}$ denotes the output of the k-th node in layer 3; $\bar{w}_k$ denotes the normalized rule weight; $\sum_{i=1}^N w_i$ represents the sum of the normalized rule weights of all fuzzy rules in the ANFIS.

4) LAYER 4: DEFUZZIFICATION LAYER

In this layer, normalized rule weights are used to generate output functions. Each node performs the following linear equation:

$$O_{4,k} = \bar{w}_k \cdot f_k = \bar{w}_k \cdot (p_k x + q_k y + r_k). \quad (24)$$

where $O_{4,k}$ denotes the output of the k-th node in layer 4; f_k denotes the contribution of the k-th rule to the overall output; p_k , q_k , and r_k are linear parameters that are adjusted during training. This equation represents the output corresponding to each rule.

5) LAYER 5: OUTPUT LAYER

This layer sums the output values from all rules to calculate the final output.

$$O_5 = \sum_{k=1}^N \bar{w}_k \cdot f_k. \quad (25)$$

where O_5 is the final output of ANFIS.

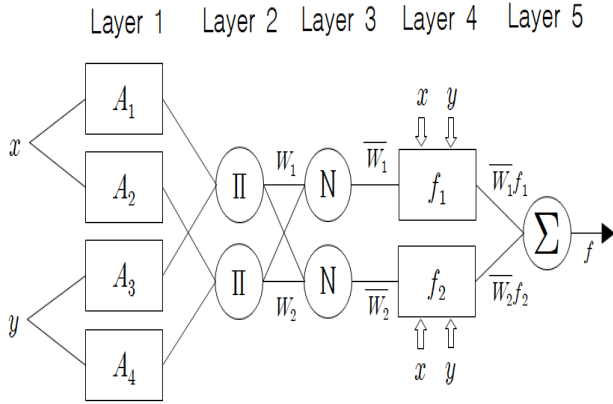


FIGURE 10. Illustration of ANFIS architecture. This figure shows the ANFIS architecture with inputs x and y and explains the ANFIS through each layer. Layer 1 converts crisp inputs into fuzzy values based on membership functions. Layer 2 applies fuzzy rules to fuzzy inputs to generate the firing strengths of each rule. Layer 3 normalizes the firing strengths to calculate the output for each rule. Layer 4 combines the outputs of all rules and applies a weighted sum to produce the final output. Layer 5 produces the final crisp output based on the weighted outputs from the previous layer.

C. ANFIS DESIGN

To improve the performance degradation of the ACC system with the fuzzy logic controller, the ANFIS is designed. The ANFIS enhances control performance through a neural network-based learning algorithm. This characteristic enables reasonable control in varying road conditions. The design process for the ANFIS is as follows.

1) TRAINING DATA OF ANFIS

The ANFIS is trained on the input-output data from the fuzzy logic controllers shown in Figure 9. The input variables of the ANFIS are θ , θ' , distance, and speed error. To overcome the limitations of the fuzzy logic controller illustrated in Figure 5, which is degrading performance under changing road conditions, the ANFIS includes θ and θ' as additional input variables. The output variable of the ANFIS is the control force. The ANFIS is designed with triangular membership functions. The triangular membership function is expressed as follows:

$$\mu_A(x) = \begin{cases} 0 & \text{if } x \leq p_l \text{ or } x \geq p_r \\ \frac{x - p_l}{p - p_l} & \text{if } p_l < x \leq p \\ \frac{p_r - x}{p_r - p} & \text{if } p < x < p_r \end{cases} \quad (26)$$

where p_l , p , and p_r represent the left endpoint, the peak, and the right endpoint of the triangle, respectively.

To improve the performance of the ANFIS, additional training data are included. Simulations are performed by varying the values of θ and θ' . Training data are obtained from simulations with θ set at -0.1 , -0.3 , and -0.5 , and θ' set at -4 . Additional training data are obtained from simulations with $\theta = -0.3$ and $\theta' = -0.7, -1.3, -1.7, -2.3, -3.7$, and

-4.3 . The ANFIS is trained on 2,000 training data points obtained from simulations.

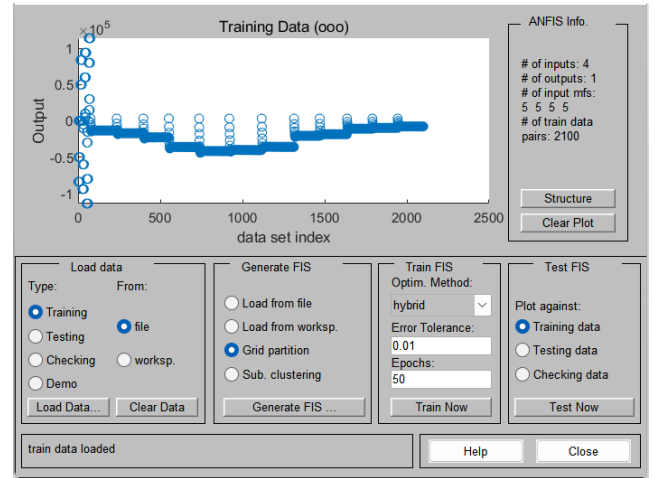


FIGURE 11. Implementation of ANFIS. This figure shows the design process of an ANFIS controller using the anfisedit function in Matlab/Simulink[®]. The training data consist of the membership function values from the fuzzy logic controller and the results from simulations performed at various angles. The optimization method is set to Hybrid Learning, the error tolerance is set to 0.01, and the number of epochs is set to 50. The inputs for the ANFIS controller are set as the inter-vehicle distance, velocity error, road angle, and rate of change of the angle.

2) ANFIS DESIGN IN MATLAB/SIMULINK[®]

The ANFIS is designed in Matlab/Simulink[®]. Figure 11 shows the design process of the ANFIS controller. The ANFIS is trained based on the three fuzzy logic controllers specifically designed for different angles, as shown in Figures 9(a), (b), and (c). These three fuzzy logic controllers shown in Figures 9(a), (b), and (c) are applied to the ACC system and simulated on the road with $\theta = -0.3$, $\theta' = -1$. The ANFIS is trained using the data obtained from these simulations. In the road environments where training data are available, the ANFIS performs comparably to the fuzzy logic controllers shown in Figures 9(a), (b), and (c). However, in road environments without training data, the ANFIS exhibits better performance than the fuzzy logic controller. The optimization method is set to Hybrid Learning [38], the error tolerance is set to 0.01, and the number of epochs is set to 50.

Figure 12 shows the application of ANFIS to the ACC system. The input variables of the ANFIS are θ , θ' , distance, and speed error. The fuzzy logic controller shown in Figure 5 performs well under specific road conditions. However, the performance of the fuzzy logic controller shown in Figure 5 degrades under changing road conditions. To overcome the limitations of the fuzzy logic controller under changing road conditions, the ANFIS incorporates θ and θ' as additional input variables. Consequently, the ANFIS demonstrates reasonable control performance even under changing road conditions.

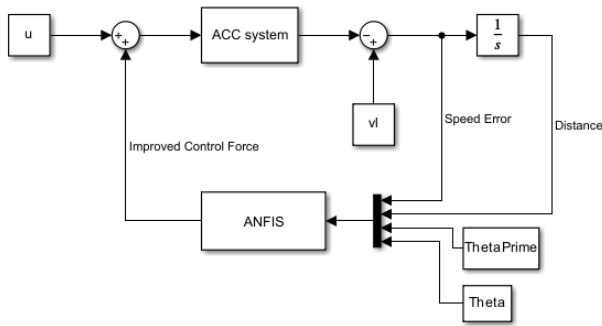


FIGURE 12. Illustration of ACC system with ANFIS controller. This figure shows the application of the ANFIS to the ACC system. The inputs of the ANFIS are θ , θ' , distance, and speed error. To ensure effective control performance in varying road conditions, θ and θ' are set as input variables. The output of the ANFIS is the control force. A rule-based system is designed through training. This rule-based system outputs the control force to enhance the performance of the ACC system in various road environments.

3) SIMULATION OF ANFIS

The simulation is performed for 60 s. To prevent the speed of the lead vehicle from becoming negative during simulation, w_l is modified as follows:

$$w_l = -200(1 + 0.01t). \quad (27)$$

A simulation is performed to verify the performance of the ANFIS trained on 2,000 data points. Figure 13 shows the performance of the ANFIS under specific road conditions. The first graph shows the performance of the ANFIS when $\theta = -0.3$ and $\theta' = -1.48$. The solid, dashed, and dotted lines represent the performance of the ANFIS, the fuzzy logic controller shown in Figure 9(a), and the fuzzy logic controller shown in Figure 5, respectively. The condition of $\theta' = -1.48$ represents a new road condition. In this case, the ANFIS exhibits better performance than both the fuzzy logic controllers shown in Figure 9(a) and Figure 5. This result shows that the ANFIS has been effectively trained using the simulation data. The second graph shows the performance of the ANFIS with $\theta = -0.314$ and $\theta' = -3.59$. In this graph, the solid, dashed, and dotted lines represent the performance of the ANFIS, the fuzzy logic controller shown in Figure 9(c), and the fuzzy logic controller shown in Figure 5, respectively. The values $\theta = -0.314$ and $\theta' = -3.59$ represent a new road condition. In this case, the ANFIS exhibits better performance than both the fuzzy logic controller shown in Figure 9(c) and the fuzzy logic controller shown in Figure 5. This represents the robustness of the ANFIS. The ANFIS learns the relationships between input and output variables through neural networks and then designs a rule-based system. Through this rule-based system, the ANFIS exhibits reasonable control performance even in unfamiliar road conditions without training data.

Figure 14 shows the distance error between the ANFIS and fuzzy logic controllers (a), (b), and (c) in Figure 9. The solid, dashed, and dotted lines represent the distance error between the ANFIS and the fuzzy logic controller shown

in Figure 9 when θ' is -1 , -2 , and -4 , respectively. When $\theta' = -1$, the distance error between the ANFIS and (a) is close to zero. This indicates that the ANFIS learns the good performance of (a) when $\theta' = -1$ through training data. When $\theta' = -2$, the distance error between the ANFIS and (b) increases compared to that when $\theta' = -1$. When $\theta' = -3$, the distance error between the ANFIS and (c) increases compared to that when $\theta' = -2$. However, because these distance errors in this figure are very small, the performance of the ANFIS remains comparable to that of (b) and (c). This indicates that the ANFIS learns the effective performance of (b) and (c) when $\theta' = -2$ and $\theta' = -4$ through training data. Overall, the ANFIS exhibits reasonable control performance under varying road conditions.

Figure 15 shows a comparison of the ACC system's performance with and without the ANFIS controller. The most substantial performance difference between the two systems occurs between 0 and 20 s, coinciding with the start of the downhill slope. This highlights the effectiveness of the ANFIS controller in regulating the vehicle's speed while descending.

To create a more realistic road scenario, two additional settings are introduced, and an additional simulation is performed. First, a road scenario with alternating uphill and downhill sections is configured, which allows the evaluation of the performance of the ANFIS in a more realistic road scenario. To configure the uphill and downhill sections, the values of θ and θ' are varied over time. θ is set to $-0.3, 0.1$, and -0.1 for the first 20 seconds, 20-40 seconds, and after 40 seconds, respectively. To ensure a smooth transition between the uphill and downhill sections, θ' is set to $-0.4, 0.2$, and -0.3 for the same time intervals. As a result, the road scenario begins with a downhill section, transitions to an uphill section in the middle, and then returns to a downhill section.

Second, the simulation is conducted with the lead vehicle's speed naturally varying according to the road gradient. The lead vehicle's speed decreases in downhill sections and increases in uphill sections. In other words, the lead vehicle's speed varies naturally according to the road gradient.

The simulation is conducted with a road scenario that is similar to real-world conditions. Figure 16 shows the distance maintained by the controlled vehicle relative to the lead vehicle on a road with alternating uphill and downhill sections. The dashed and solid lines represent the performance of the conventional ACC system and the ACC system with the ANFIS, respectively. The first 20 seconds and the period after 40 seconds correspond to downhill sections. Generally, when a vehicle enters a downhill road, its speed increases. Therefore, in this simulation, a sufficient distance is maintained between the two vehicles to ensure a stable following distance. The 40-60 second interval represents an uphill section. Typically, when a vehicle enters an uphill road, its speed increases to climb the slope, temporarily reducing the gap between the vehicles. Compared with the conventional ACC system, the ACC system with the

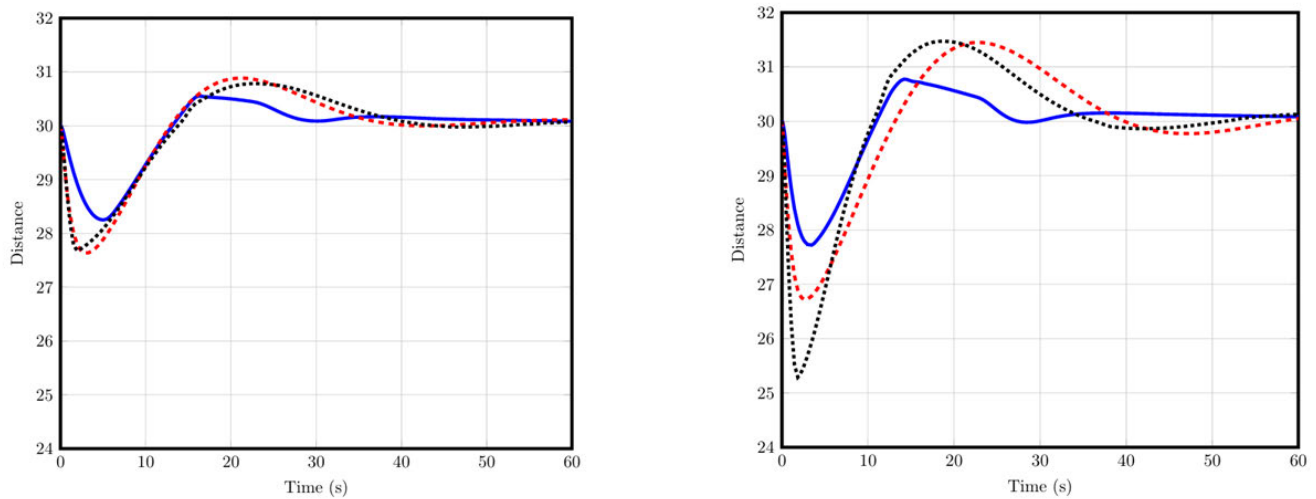


FIGURE 13. Graphs of the ANFIS controller. This figure shows the performance of the ANFIS under specific road conditions. The first graph shows the performance of the ANFIS when $\theta = -0.3$ and $\theta' = -1.48$. The solid, dashed, and dotted lines represent the performance of the ANFIS, the fuzzy logic controller shown in Figure 9(a), and the fuzzy logic controller shown in Figure 5, respectively. The ANFIS does not have data for $\theta' = -1.48$. However, the ANFIS exhibits superior performance compared to the fuzzy logic controller. The second graph shows the performance of the ANFIS when $\theta = -0.314$ and $\theta' = -3.59$. The solid, dashed, and dotted lines represent the performance of the ANFIS, the fuzzy logic controller shown in Figure 9(c), and the fuzzy logic controller shown in Figure 5, respectively. The ANFIS lacks data for $\theta = -0.314$ and $\theta' = -3.59$. However, the ANFIS exhibits superior performance compared to the fuzzy logic controller. This result demonstrates that the ANFIS exhibits good performance under new road conditions by learning using a neural network.

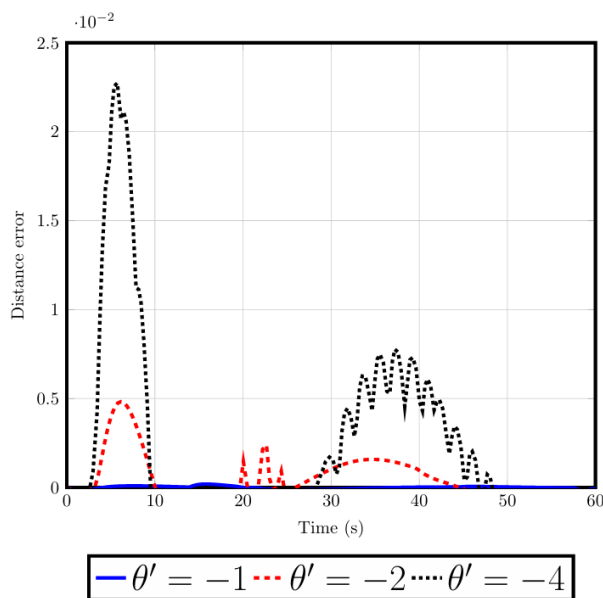


FIGURE 14. Distance error between ANFIS and Figures 9(a), (b), and (c). This figure shows the difference in simulation results when ANFIS is applied to the ACC system and compared to the fuzzy logic controllers in Figures 9. The solid, dashed, and dotted lines represent the distance errors between ANFIS and Figures 9(a), (b), and (c) when $\theta' = -1$, -2 , and -4 , respectively. For $\theta' = -1$, -2 , and -3 , the distance error between the ANFIS and Figure 9(a), (b), and (c) are small, indicating that the performance of the ANFIS is comparable to that of the controllers in Figures 9(a), (b), and (c). This figure shows that ANFIS exhibits the same performance as the fuzzy logic controller under specific road conditions.

ANFIS maintains a distance closer to the reference input value. This reduces the risk of collision with the lead

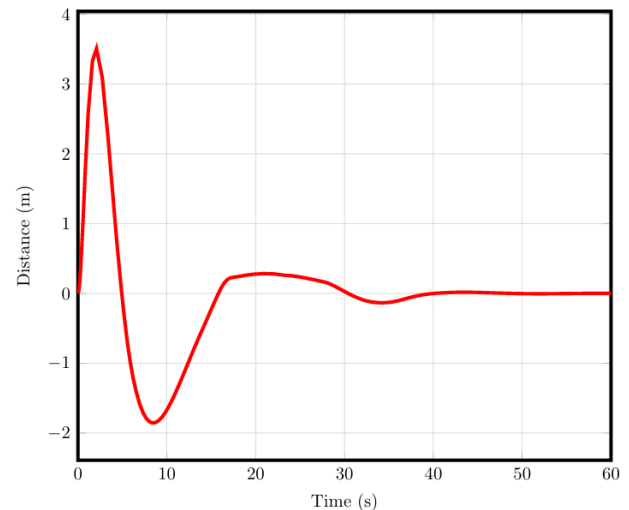


FIGURE 15. Performance error of the ACC system with and without the ANFIS controller. This figure shows a graph that highlights the performance difference between the ACC system without the ANFIS controller and the one using it. The most substantial performance error between the two systems occurs between 0 and 20 seconds. This is the section where the ANFIS controller regulates the speed of the vehicle descending a downhill slope. The graph clearly demonstrates that the performance of the ACC system has been significantly improved by the ANFIS controller.

vehicle and enhances the stability of the ACC system. In addition, the conventional system exhibits abrupt changes in the intervehicle distance when the road gradient shifts. In contrast, the ACC system with the ANFIS demonstrates smoother distance adjustments based on the simulation data, thereby improving passenger comfort. In summary, the

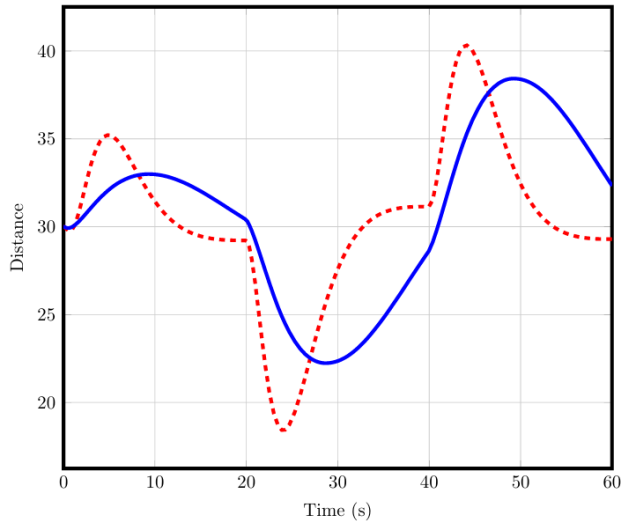


FIGURE 16. Simulation of road conditions with uphill and downhill sections and varying lead vehicle speed. This figure compares the performance of a conventional ACC system and an ACC system with the ANFIS in a road environment that simulates real-world conditions. The graph shows the distance maintained by the controlled vehicle relative to the lead vehicle over time. The dashed and solid lines represent the distance maintained by the conventional ACC system and the ACC system with the ANFIS, respectively. The first 20 s and the period after 40 s correspond to downhill sections, and the 20–40 s interval represents an uphill section. On downhill roads, where vehicle speed increases, the controlled vehicle reduces its speed to maintain a safe distance from the lead vehicle. Conversely, on uphill roads, where vehicle speed decreases, the controlled vehicle increases its speed to climb the slope. The ACC system with the ANFIS maintains a distance that is closer to the reference input than the conventional system. While the conventional system exhibits abrupt speed changes when the road gradient shifts, the ANFIS-based system demonstrates smoother speed adjustments. These results indicate that the ACC system with ANFIS outperforms the conventional system in maintaining stability and control.

simulation results confirm that the ACC system with the ANFIS outperforms the conventional system.

V. CONCLUSION

This study enhances the performance of ACC systems on sloped roads using fuzzy logic control and the ANFIS. First, an ACC system was implemented that assumes a downhill road. The fuzzy logic controller was then designed by establishing input-output membership functions and a rule-based system. The fuzzy logic controller was applied to the ACC system, and the improved performance of the ACC system was verified via simulations. However, the fuzzy logic controller faced performance degradation in varying road conditions. To overcome this limitation, the ANFIS was proposed. The ANFIS was trained on fuzzy logic controller simulation data and data specifically designed for various road angles. The trained ANFIS controller is applied to the ACC system, and simulations are performed at specific road angles. The ANFIS maintained the enhanced performance of the fuzzy logic controller designed for specific angles. Even at new road angles, the ANFIS exhibited reasonable control performance under changing road conditions. To the best knowledge of the authors, applying the ANFIS to the ACC system is the first approach in this context.

In this study, a linearized model of vehicle dynamics was used to evaluate the performance of the ACC system. In future research, it is necessary to replace the linear model with a nonlinear dynamics model to enable simulations that can be applied to real vehicles. This study focuses on analyzing the interactions between lead and controlled vehicles. However, considering more than two vehicles is more realistic. To simulate real-world conditions, it is essential to consider situations where multiple vehicles interact on the road. In this process, it is necessary to consider a collision avoidance algorithm. The performance of the ACC system should be analyzed by considering interactions between multiple vehicles, thereby enhancing the feasibility and reliability of ACC systems. In future research, the offline data used in the current study is adjusted to values that are suitable for real road environments. Then, the ANFIS is trained on these data and applied to a vehicle to simulate its actual performance. The results obtained from this simulation are compared with those of the existing system and online simulations to verify the superiority of the ANFIS controller. This process is expected to serve as an appropriate method to ensure a fair comparison of the system's performance.

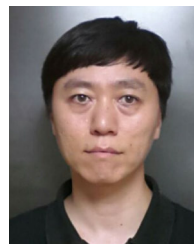
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